6

Brain Rhythms Provide a Framework for Neural Syntax

> Rhythm imposes unanimity upon the divergent, melody imposes continuity upon the disjointed, and harmony imposes compatibility upon the incongruous.
>
> —Yehudi Menuhin[1]

> Language is a process of free creation; its laws and principles are fixed, but the manner in which the principles of generation are used is free and infinitely varied.
>
> —Noam Chomsky[2]

> I would define, in brief, the poetry of words as the rhythmical creation of beauty.
>
> —Edgar Allan Poe[3]

I was mesmerized when, as a middle school student in Hungary, I first heard Morse code in a ham radio course I attended. Sending signals through the ether with a transmitter, the instructor told us that somebody from New Zealand had responded to his call. The operator on the other side of the world informed us about the sunny weather there, the receiving conditions, the details of his rig, and the antenna type he used. My surprise only increased

1. https://en.wikiquote.org/wiki/Yehudi_Menuhin.
2. http://www.famousphilosophers.org/noam-chomsky/.
3. Poe, EA (The Poetic Principle; CreateSpace Publishing, 2016).

The Brain from Inside Out. György Buzsáki, Oxford University Press (2019).
 DOI: 10.1093/oso/9780190905385.003.0006

when I figured out that our instructor did not speak either English or Maori, and presumably the operator in New Zealand did not understand Hungarian. Instead, I was informed, they "spoke" to each other with dots (short pulses) and dashes (long pulses), through a universal language in which all words consist of three letters, called *Q-code*. To be part of such a conversation, all I had to do was to learn Morse code, memorize the Q language, learn a bit about electronics, pass exams, get a license, build a transmitter and receiver, and set up a wire antenna between the chimneys of our house and the neighbor's. Then I could communicate with any ham radio around the globe. That is exactly what I did, and the problem of coding has bugged me ever since.

In learning Morse code, the biggest hurdle is to recognize letters in the sea of dots and dashes separated by short spaces. Separation of messages is the most important thing in any coding system, be it speech, written language, computer language, or spike transmission in the brain. In Morse code, letters are separated by silent periods of the duration of a dot, whereas words have boundaries of at least two dots' length: [.. _ _ _ . ._ _._ . . . . _ . . . .] contains information (translation: "it makes sense"), whereas [..___..__._. . . .. . . ._....] the exact same sequence of dots and dashes, but without spaces between the characters, is just nonsense noise. Similarly, if this entire page were a single long word, it would be very difficult to decipher the content, although not impossible because unique combinations of thirty letters are much easier to decode than combinations of only two characters. The situation with neuronal messages is even more complicated because action potentials are all pretty much the same: just dots.[4]

Human language is also produced by combining elements. For instance, linking letters in particular sequences produces words, which have new qualities not present in the individual letters. Such combinations are regulated by *syntax*, a set of rules that govern the transformation and temporal progression of discrete elements (such as letters or musical notes) into ordered and hierarchical relations (words, phrases, and sentences, or chords and chord progressions) that allow the brain to interpret their meaning. Grouping or

4. In principle, single spikes and bursts of spikes could be conceived of as dots and dashes. Indeed, it has been suggested that the postsynaptic neuron can effectively discriminate between single spikes and burst events (Lisman, 1997). Ham radio operators are not exceptional among neuroscientists. I have met a few of them, including the late W. Ross Adey (UCLA), John Hopfield (Princeton University), Terry Sejnowski (Salk Institute), and Fritz Sommer (UC Berkeley). I also had radio contacts with a few famous people, including Yuri Gagarin (UA1LO), Senator Barry Goldwater (K7UGA), King Hussein of Jordan (JY1), and his Queen Noor (JY1H). The aviator Howard Hughes was also a ham radio operator (W5CY). Q-code is also used in maritime communication. In addition to Q-code, hams use a more specialized code, which includes some arbitrary signals (e.g., 88 = love and kisses), while others are simplified English words (e.g., vy = very, gm = good morning, r = are).

chunking the fundamentals by syntax[5] allows the generation of virtually infinite numbers of combinations from a finite number of lexical elements using a minimal number of rules in music, human, sign, body, artificial, and computer languages, as well as in mathematical logic.[6] With just thirty or so letters (or sounds) and approximately 40,000 words, we can effectively communicate the entire knowledge of humankind. This is an extraordinary combinatorial feat. Syntax is exploited in all systems where information is coded, transmitted, and decoded.[7] In this chapter, I suggest that neuronal rhythms provide the necessary syntactical rules for the brain so that unbounded combinatorial information can be generated from spike patterns.

A HIERARCHICAL SYSTEM OF BRAIN RHYTHMS: A FRAMEWORK FOR NEURAL SYNTAX?

Researchers who study neuronal rhythms or oscillations have created an interdisciplinary platform that cuts across psychophysics, cognitive psychology, neuroscience, biophysics, computational modeling, and philosophy.[8] Brain rhythms are important because they form a hierarchical system that offers a

5. Segmentation or chunking has been often considered as a basic perceptual and memory operation. According to Newtson et al. (1977), segmentation is an interplay between perceptual, bottom-up processes, and inferential, top-down processes. The observer has a preformed anticipatory schema that helps to select features that are characteristic for anticipated events or activities and their change. Recognition of event boundaries partitions a continuous stream of events into discrete units and forms the basis for a redefinition of the search and extraction of new information. The "preformed anticipatory schema" may correspond to the system of brain rhythms. See also Shipley and Zacks (2008).

6. Several previous thinkers have considered the need for brain rules to support syntax in language; for example Port and Van Gelder (1995); Wickelgren (1999); Pulvermüller (2003, 2010). Karl Lashley (1951) was perhaps the first who specifically thought that sequential and hierarchical neuronal organization must be behind "fixed action patterns" or "action syntax," behaviors that are elicited by ethologically relevant cues but proceed according to species-specific rules even without prior experience (Tinbergen, 1951).

7. In a strict sense, "syntax" (from Greek *syntaxis)* means "arrangement," and in language it roughly refers to the ordering of words (i.e., what goes where in a sentence). It does not provide any information about meaning, which is the domain of semantics. Sometimes syntax is used synonymously with *grammar*, although grammar is a larger entity that includes syntax. Grammar sets the rules for how words are conjugated and declined according to aspect, tense, number, and gender.

8. I devoted a book (*Rhythms of the Brain*, 2006), a monograph (Buzsáki, 2015), and several reviews (e.g., Buzsáki, 2002; Buzsáki and Draguhn, 2004; Buzsáki and Wang, 2012) to this topic. See also Whittington et al. (2000), Wang (2010); Engel et al. (2001); Fries et al. (2001); Fries (2005).

syntactical structure for the spike traffic within and across neuronal circuits at multiple time scales, and their alterations invariably lead to mental and neurological disease.[9]

The transmembrane potential of neurons can be measured by electrodes, which can report neuronal activity with submillisecond time resolution. Electric current flows from many neurons superimpose at a given location in the extracellular medium to generate a potential with respect to a reference site. We measure the difference between "active" and reference locations as voltage. When an electrode with a small tip is used to monitor the extracellular voltage contributed by hundreds to thousands of neurons, we refer to this signal as the *local field potential* (LFP). [10] Recordings known as the *electrocorticogram* (ECoG) are made with larger electrodes placed on the brain surface or its surrounding dura mater to sample many more neurons. Finally, when even larger footprint electrodes are placed on the scalp, we call the recorded signal an *electroencephalogram* (EEG). Each measure refers to the same underlying mechanisms and neuron voltages, but because of the different sizes of the monitoring electrodes used in LFP, ECoG, and EEG recordings, they integrate over increasingly larger numbers of neurons, from a few dozens to millions. The magnetic field induced by the same transmembrane activity of neurons is known as the *magnetoencephalogram* (MEG).[11]

The patterns measured by LFP, ECoG, EEG, or MEG are very useful for the experimenter because they provide critical information about the timing of cooperating neurons. Like noise measured in a football stadium, these techniques cannot provide information about individual conversations but

9. In the brain (and other systems), hierarchy is defined as an asymmetrical relationship between forward and backward connections.

10. The term LFP is an unfortunate misnomer, since there is no such a thing as "field potential." What the electrode measures is voltage in the extracellular space (*V*e; a scalar measured in volts). The electric field is defined as the negative spatial gradient of *Ve*. It is a vector whose amplitude is measured in volts/meter. All transmembrane ionic currents, from fast action potentials to the slowest voltage fluctuations in glia, at any given point in brain tissue, superimpose to yield *V*e at that location. Any excitable membrane (spine, dendrite, soma, axon, and axon terminal of neurons) and any type of transmembrane current contributes, including slow events in glia to pericytes of capillaries. *V*e amplitude scales with the inverse of the distance between the source and the recording site. The LFP and EEG waveforms, typically measured as amplitude and frequency, depend on the proportional contribution of the multiple sources and various properties of brain tissue. A major advantage of extracellular field recording is that, in contrast to other methods used to investigate network activity, the biophysics related to these measurements are well understood (Buzsáki et al., 2012; Einevoll et al., 2013).

11. Hämäläinen et al. (1993).

can precisely determine the timing of potentially important events, such a goal in the stadium or the emergence of a synchronized pattern in the brain. Monitoring LFPs or magnetic fields is the best available method for identifying brain state changes and faithfully tracking dynamic population patterns such as brain rhythms.

A System of Rhythms

There are numerous brain rhythms, from approximately 0.02 to 600 cycles per second (Hz), covering more than four orders of temporal magnitude (Figure 6.1).[12] Many of these discrete brain rhythms have been known for decades, but it was only recently recognized that these oscillation bands form a geometric progression on a linear frequency scale or a linear progression on a natural logarithmic scale, leading to a natural separation of at least ten frequency bands.[13] The neighboring bands have a roughly constant ratio of e = 2.718—the base for the natural logarithm.[14] Because of this non-integer relationship among the various brain rhythms, the different frequencies can never perfectly entrain each other. Instead, the interference they produce gives rise to metastability, a perpetual fluctuation between unstable and transiently stable states, like waves in the ocean. The constantly interfering network rhythms can never settle to a stable attractor, using the parlance of nonlinear dynamics. This explains the ever-changing landscape of the EEG.

12. Rhythms are a ubiquitous phenomenon in nervous systems across all phyla and are generated by devoted mechanisms. In simple systems, neurons are often endowed with pacemaker currents, which favor rhythmic activity and resonance in specific frequency bands (Grillner, 2006; Marder and Rehm, 2005).

13. Penttonen and Buzsáki (2003); Buzsáki and Draguhn (2004). The multiple time scale organization of brain rhythms is similar to Indian classical music, where the multilevel nested rhythmic structure, explained by the concept of *tāla*, characterizes the composition. *Tāla* roughly corresponds to a rhythmic framework and defines a broad structure for repetition of musical phrases, motifs, and improvisations. Each *tāla* has a distinct cycle in which multiple other faster rhythms are nested (Srinivasamurthy et al., 2012).

14. The constant e is sometimes called Napier's constant (see Chapter 12) and is perhaps the most famous irrational number with the exception of π. Yet the e symbol honors the German mathematician Leonhard Euler who defined it. If you ever wondered why naming things and getting credit for discoveries are often an idiosyncrasy of history, read these entertaining accounts about credit assignment debates in mathematics: Maor (1994); Conway and Guy (1996); Beckmann (1971); Livio (2002); Posamentier and Lehmann (2007).

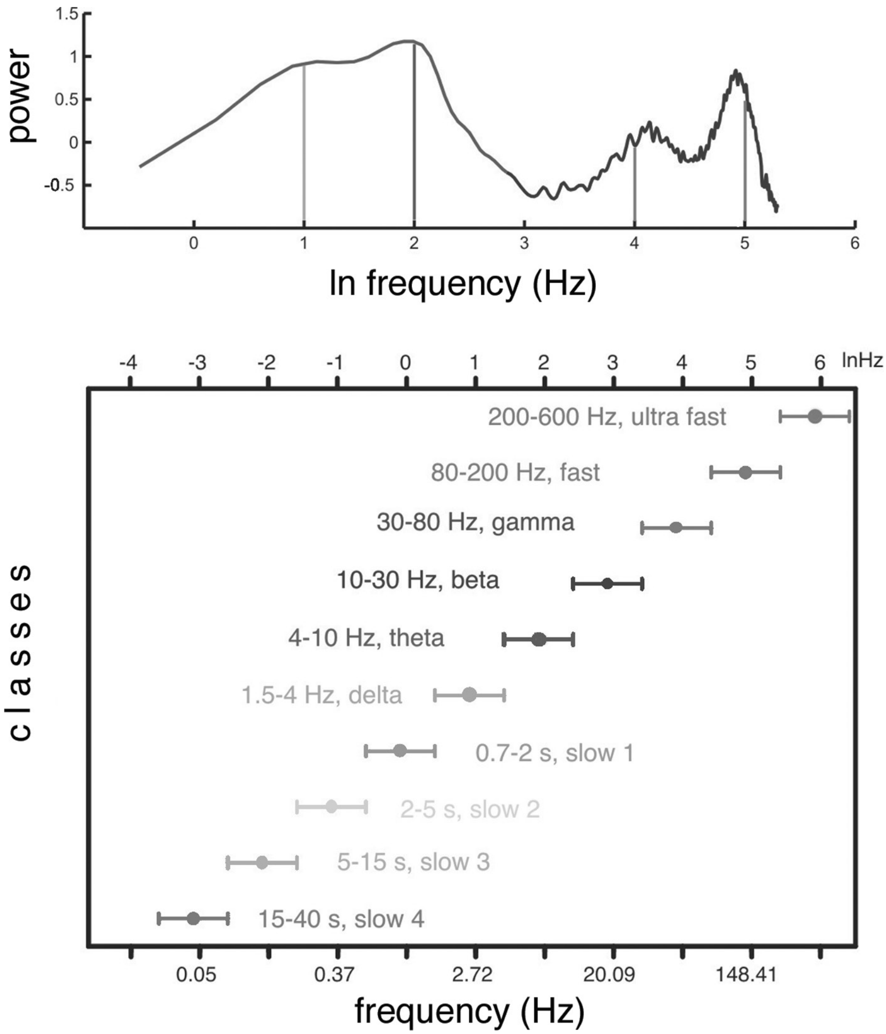

Figure 6.1. Multiple oscillators form a hierarchical system in the cerebral cortex. *Top*: Power spectrum of hippocampal electroencephalogram (EEG) in the mouse recorded during sleep and waking periods. Note that the four peaks (1, 2, 4, 5), corresponding to the traditional delta, theta, gamma, and fast ("ripple") bands, are multiples of natural log integer values (ln frequency). *Bottom*: Oscillatory classes in the cerebral cortex show a linear progression of the frequency classes on the log (ln) scale. In each class, the frequency ranges ("bandwidth") overlap with those of the neighboring classes so that frequency coverage is more than four orders of magnitude.
Modified with permission from Penttonen and Buzsáki (2003).

Traditionally, the frequency bands are denoted by Greek letters.[15] Several rhythms can temporally coexist in the same or different structures and interact with each other. Their constellation determines various brain states, such as stages of sleep and the arousal levels of waking. The different oscillations generated in cortical networks show a hierarchical relationship, often expressed by *cross-frequency phase modulation* between the various rhythms. This term implies that the phase of a slow oscillation modulates the amplitude of a faster

15. The Greek letters do not indicate a logical order of frequencies but roughly the order of their discoveries. Hans Berger (1873–1941), the discoverer of the human EEG, first observed 8–12 Hz rhythmic patterns above the occipital cortical area when his study participants closed their eyes, and he called them alpha waves (Berger, 1929). In their absence, the smaller-amplitude, faster vibrations, present when the eyes were open, were named beta waves (13–30 Hz). The largest amplitude waves are present during deep sleep and are known as delta waves (0.5–4 Hz). We already discussed theta oscillations (4–10 Hz), most prominent in the hippocampus. The story of gamma oscillations is really fascinating. Perhaps Jasper and Andrews (1938) used the term "gamma waves" first for frequencies between 35 and 45 Hz. The idea that this "40-Hz" oscillation is a "cognitive" rhythm likely originates from Henri Gastaut (Das and Gastaut, 1955), who described 40-Hz rhythmic trains in the scalp EEG of trained yogis during the *samadhi* state. Banquet (1973) also observed 40-Hz bouts during the third deep stage of transcendental meditation. In normal participants, Giannitrapani (1966) found increases in 35–45 Hz immediately before they answered difficult multiplication questions. Subsequently, Daniel Sheer popularized "40 Hz" in his many papers on biofeedback (e.g., Sheer et al., 1966; Bird et al., 1978), which he thought was basically a high frequency "beta" mechanism. The phrase "gamma rhythm" became popular in the 1980s. According to the late Walter Freeman "I coined it in 1980, when the popular term was '40 Hz'. . . . At that time Steve Bressler was beginning graduate work in my lab, so I assigned him to a literature search to document the inverse relation of OB [olfactory bulb] frequency and size. Berger had coined alpha. Someone else coined beta (I don't remember who), and Berger coopted it. In analogy to particle physics, the next step up would be gamma. I found no prior use of 'gamma' in EEG research, so Steve and I called it that. We sent the manuscript to Mollie Brazier, long-time editor of the *EEG Journal*. She wrote that she would submit our term to the Nomenclature of the International EEG Society. A month later she reported back that the committee had refused to endorse the usage, so she wouldn't publish the paper unless we took it out. Steve needed a publication to get support, so we complied. When the article appeared (Bressler and Freeman, 1980), there it was: 'Gamma rhythms in the EEG,' in the running title. I'd forgotten to take it out, and nobody noticed, least of all Mollie. So that's how it first appeared in print. I continued to use it in lectures, insisting that gamma is a range, not a frequency, and it caught on. Now, like numerous successful developments in science, few people know where it came from" (cited from my e-mail correspondence with WF, April 3, 2011). Subsequently, Steve Bressler sent me a copy of his original manuscript (he kept it for 30 years!). Voila, its title reads "EEG Gamma Waves: Frequency Analysis of Olfactory System EEG in Cat, Rabbit, and Rat." In the Introduction, they acknowledge that "The label 'gamma waves' was used by Jasper and Andrews (1938)," but Walter did not remember this detail. Putting aside the unavoidable errors of source memory, we should thank Walter Freeman for reintroducing gamma oscillations. *Gamma rhythm* catapulted to its current fame after Wolf Singer and his colleagues suggested that it could be the Rosetta Stone for the problem of perceptual binding (Gray et al., 1989).

rhythm, meaning that its amplitude varies predictably within each cycle. In turn, the phase of the faster rhythm modulates the amplitude of an even faster one and so on. This hierarchical mechanism is not unique to the brain. Spring, summer, fall, and winter are four phases of a year that "modulate" both the amplitude and duration of day length. In turn, the phases of the day are correlated (i.e., phase-locked) with the alignment of the sun and moon, which modulate the tidal magnitude of the oceans.

Because of cross-frequency coupling, the duration of the faster event is limited by the "allowable" phase extent of the slower event. The fastest oscillation is phase-locked to the spikes of the local neurons and, because of cross-frequency coupling, to all slower rhythms of the hierarchy. For example, the ultrafast oscillatory "ripple" waves (5–7 ms or ~150–200/s) in the hippocampus are phase-locked to the spikes of both pyramidal cells and several types of inhibitory interneurons, and the magnitude of the short-duration ripple events (approximately 40–80 ms) is modulated by the phase of thalamocortical sleep spindles. The spindle events, in turn, are phase-modulated by cortical delta oscillations, which are nested in the brain-wide infra-slow (slow 3) oscillations (Figure 6.2).[16] The nested nature of brain rhythms may represent the needed structure for syntactic rules, allowing for both the separation of messages (e.g., gamma oscillation cycles containing cell assemblies; see Chapter 4) and their linking into neuronal words and sentences.

PUNCTUATION BY INHIBITION

The excitatory pyramidal neurons (also called principal cells) are considered to be the main carriers of information in the cortex. Their potential runaway excitation is curtailed by inhibitory interneurons: the 15–20% of cortical neurons that contain the inhibitory neurotransmitter gamma aminobutyric acid (GABA). The main function of these neurons is to coordinate the flow of excitation in neuronal networks. There are several different classes of inhibitory

16. Sirota et al. (2003). There are many examples of cross-frequency coupling across structures and species (Buzsáki et al., 1983; Soltesz and Deschênes, 1993; Steriade et al., 1993a, 1993b, 1993c; Sanchez-Vives et al., 2000; Bragin et al., 1995; Buzsáki et al., 2003; Csicsvari et al., 2003; Chrobak and Buzsáki, 1998; Leopold et al., 2003; Lakatos et al., 2005; Canolty et al., 2006; Isomura et al., 2006; Sirota et al., 2008). In fact, nobody has ever seen an exception. For excellent reviews, see Jensen and Colgin (2007); Axmacher et al. (2010); Canolty and Knight (2010). Yet caution should be used in the quantification of cross-frequency phase coupling as waveform distortion and nonstationarities can often yield spurious coupling (Aru et al., 2015). One way to avoid artifactual coupling is to first establish that two independent oscillations exist in the examined network.

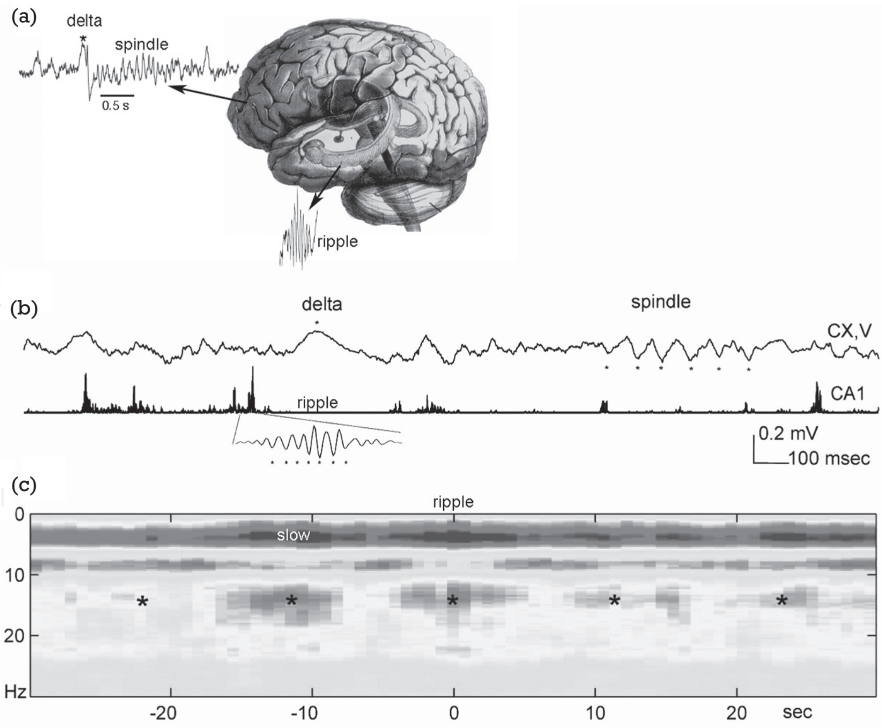

Figure 6.2. Hippocampal sharp wave ripples, neocortical slow waves and sleep spindles are often temporally coupled. A: Schematics of events and structures. B: Example traces of neocortical (CX, V) local field potential and power of fast oscillations (100-200 Hz) in the hippocampal (CA1) local field potential in the rat. An example of the fast local oscillation (ripple) is also shown. C: Hippocampal ripple peak-triggered (time zero) neocortical spectrogram. Note increased correlation of power in the slow oscillation (slow) and sleep spindle (10–18 Hz) bands with hippocampal ripples. *Ultra-slow (~0.1 Hz) comodulation of neocortical and hippocampal activity.
Reproduced from Sirota et al. (2003).

interneurons with specialized functions. One group selectively innervates the axon's initial segment of pyramidal cells, where the action potential is produced, and can influence the timing of spikes. A second group innervates the region surrounding the cell body and can electrically segregate the dendrites from the axon. A third group is dedicated to dendritic inhibition; their main function is to attenuate or "filter out" excitatory inputs onto different dendritic segments. Dendrite-targeting interneurons can also effectively short-circuit entire dendritic branches or trees and thus dynamically alter the biophysical properties of the inhibited principal neurons. Yet another group exclusively inhibits other interneurons. Interneurons collect excitatory inputs from the surrounding pyramidal cells and respond equally effectively to distant excitatory neurons or to subcortical neurons carrying various neuromodulators. In turn, the axons of most interneurons arborize locally, providing inhibition to the surrounding pyramidal cell population. Finally, a smaller important group of inhibitory cells project their axons to distant structures, hence their name: *long-range interneurons*.

Although there is no agreed job description for each of these interneuron types, their overall task is akin to traffic controllers in a big city. The ability to stop or slow excitation and route the excitatory traffic in the desired direction is an important requirement in complex networks. To be effective, the various traffic controllers should be temporally well-coordinated for each given job.[17] Inhibition of excitatory neurons can be conceived as the punctuation marks of a neural syntax that can parse and segregate neuronal messages.

The segregating or gating effect of neuronal oscillations can be illustrated by considering a single neuron whose membrane potential is fluctuating around the action potential threshold. The outcome of afferent excitation of a neuron depends on the state of the neuron. If the membrane potential is close to the threshold, a very small amount of excitation is enough to discharge the cell. However, when afferent excitation arrives at the time of hyperpolarization (i.e., when the membrane potential is more negative than at rest), the input may be ignored. Because axons of the interneurons target many principal cells, inhibition can effectively synchronize the action of the principal cells. If the discharge of interneurons is temporally coordinated—for example, by oscillatory mechanisms—many pyramidal cells in the network can produce synchronous output and exert a stronger effect on their downstream targets compared with their noncoordinated or asynchronous firing. In sum, the concerted action of interneurons can route the excitatory information at the right time and in the right direction.

Inhibition Creates Oscillations

Inhibition is the foundation of brain rhythms, and every known neuronal oscillator has an inhibitory component. Balance between opposing forces,

17. There are several reviews on cortical interneurons (Freund and Buzsáki, 1996; Buzsáki et al., 2004; McBain and Fisahn, 2001; Soltesz, 2005; Rudy et al., 2011; Klausberger and Somogyi, 2008; DeFelipe et al., 2013; Kepecs and Fishell, 2014). Long-range interneurons are also found in the cortex and may be related to other inhibitory neuron types with long axons, such as the medium spiny neurons of the striatum and GABAergic neurons of the ventral tegmental area, substantia nigra, raphe nuclei, and several other brainstem areas.

such as excitation and inhibition, can be achieved most efficiently through oscillations. The output phase allows the transmission of excitatory messages, which are then gated by the build-up of inhibition. Oscillations are energetically the cheapest way to synchronize the constituents of any system, be it mechanical or biological. This may be a good reason for brain networks to generate oscillations spontaneously when no useful external timing signals are available. Inhibitory interneurons can act on many target neurons synchronously, effectively creating windows of opportunity for afferent inputs to affect inhibition-coordinated local circuits. In summary, oscillatory timing can transform both interconnected and unconnected principal cell groups into transient coalitions, thus providing flexibility and economical use of spikes.

BRAIN-WIDE COORDINATION OF ACTIVITY BY OSCILLATIONS

In contrast to computers, where communication is equally fast between neighbors and physically distant components, axon conduction delays in the brain limit the recruitment of neurons into fast oscillatory cycles. As a result, infra-slow waves are of tsunami size and involve many neurons in large brain areas, whereas superfast oscillations are just ripples riding on the vortex of the slower waves and contribute mostly to local integration. The consequence of such cross-frequency relationships is that perturbations at slow frequencies inevitably affect all nested oscillations.

An important utility of cross-frequency phase coupling of rhythms is that the brain can integrate many distributed local processes into globally ordered states. Local computations and the flow of multiple signals to downstream reader mechanisms can be brought under the control of more global brain activity, a mechanism usually referred to in cognitive sciences as *executive, attentional, contextual*, or *top-down control*.[18] These terms reflect the tacit recognition that sensory inputs alone are not enough to account for the variability of network activity, so some other sources of drive must be assumed. These other sources can be conceived of as a supervising signal coming from preexisting wiring and the brain's knowledge base, which can provide the necessary grounding for the meaning of the sensory input (Chapter 3).

Global coordination of local computations in multiple brain areas through cross-frequency coupling can ensure that information from numerous areas

18. Engel et al. (2001); Varela et al. (2001).

is delivered within the integration time window of downstream reader mechanisms. A useful analogy here is the cocktail party problem, when many people speak at the same time. Comprehending a conversation is much easier when the listener can also see the actions of the speaker. Movements of the lips and facial muscles involved in speech production precede the sound by several tens of milliseconds, and their patterns are characteristically correlated with the uttered sounds. Thus, visual perception of the sound production information can prepare and enhance sound perception. Visual information reaches the auditory areas of the brain just in time to be combined with the arriving auditory input. Indeed, watching a silent movie with human speakers activates auditory areas of the listener's brain.[19] Understanding someone with a foreign accent is difficult initially, but the listener can catch up quickly by combining auditory and visual information. A temporal discrepancy between acoustic and visual streams may introduce comprehension difficulties, but people quickly learn to compensate for it.

Input–Output Coordination

Neuronal oscillations have a dual function in neuronal networks: they influence both input and output neurons. Within oscillatory waves, there are times when responsiveness to a stimulus is enhanced or suppressed. We can call them "ideal" and "bad" phases. Oscillation is an energy-efficient solution for periodically elevating the membrane potential close to threshold, thus providing discrete windows of opportunity for the neuron to respond. The physiological explanation for this gating effect is that the bad phase of the oscillatory waves is dominated by inhibition, as discussed earlier, whereas excitation prevails at the ideal phase. The same principle applies at the network level: when inputs arrive at the ideal phase of the oscillation—that is, at times when neurons fire synchronously and thus send messages, they are much less effective compared to the same input arriving during the bad phase of the oscillator, when most neurons are silent.

As discussed earlier, activation of inhibitory interneurons can hyperpolarize many principal neurons simultaneously. Recruitment of inhibitory interneurons can happen via either afferent inputs (feedforward) or via pyramidal neurons of the activated local circuit (feedback). As a result, the same mechanism that gates the impact of input excitation also affects the timing of output spikes in

19. Schroeder et al. (2008); Schroeder and Lakatos (2009).

many neurons in the local circuit.[20] Such synchronized cell assembly activity can have a much larger impact on downstream partners than on individual, uncoordinated neurons with irregular interspike intervals. This dual function of neuronal oscillations is what makes them a useful mechanism for chunking information into packages of various lengths.[21]

BRAIN RHYTHMS ACROSS SPECIES

The brain is among the most sophisticated scalable architectures in nature. Scalability is a property that allows a system to grow while performing the same computations, often with increased efficiency. In scalable architectures, certain key aspects of the system should be conserved to maintain the desired function while other aspects compensate for such conservation.

Temporal organization of neuronal activity, as represented by rhythms, is a fundamental constraint that needs to be preserved when scaling brain size. Indeed, perhaps the most remarkable aspect of brain rhythms is their evolutionarily conserved nature. Every known pattern of LFPs, oscillatory or intermittent, in one mammalian species is also found in virtually all other mammals investigated to date. Not only the frequency bands but also the temporal aspects of oscillatory activity (such as duration and temporal evolution) and, importantly, their cross-frequency coupling relationships and behavioral correlations, are also conserved (Figure 6.3). For example, the frequency, duration, waveform,

20. The idea that randomly occurring inputs will produce outputs determined by the phase of the oscillating target networks can be traced to Bishop (1933). Bishop stimulated the optic nerve and observed how the cortical response varied as the function of peaks and valleys of the EEG. Many subsequent models of oscillatory communication are based on these observations (e.g., Buzsáki and Chrobak, 1995; Fries, 2005).

21. Brain rhythms belong to the family of weakly chaotic oscillators and share features of both harmonic and relaxation oscillators. The macroscopic appearance of several brain rhythms resembles the sinusoid pattern of harmonic oscillators, although the waves are hardly ever symmetric. An advantage of harmonic oscillators is that their long-term behavior can be reliably predicted from short-term observations of their phase angle. If you know the phase of the moon today, you can calculate its phase a hundred years from now with high precision. This precision of the harmonic oscillators is also a disadvantage, as they are hard to perturb and, therefore, they poorly synchronize their phases. Functionally, neuronal oscillators behave like relaxation oscillators, with a "duty phase" when spiking information is transferred, followed by a refractory period. This refractory phase of the cycle can be called the perturbation or "receiving" phase because in this phase, the oscillator is "vulnerable," and its phase can be reset. Due to the separation of the sending and receiving phases, relaxation oscillators can synchronize robustly and rapidly in a single cycle, making them ideal for packaging spiking information in both time and space.

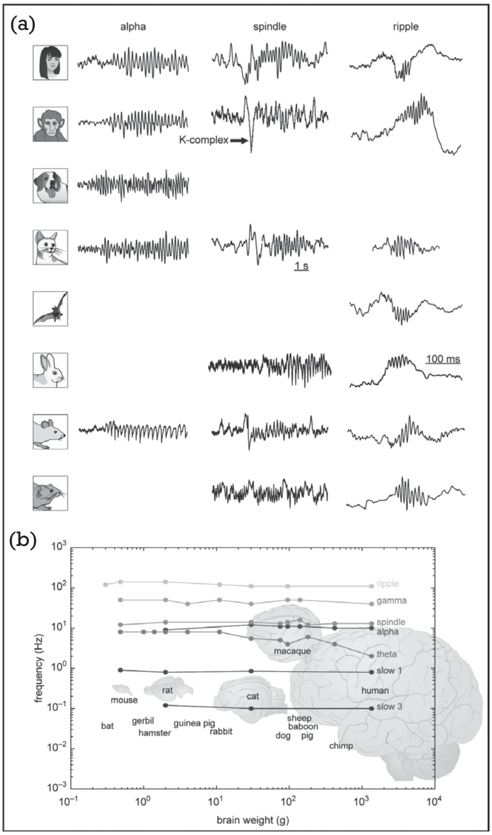

Figure 6.3. Preservation of brain rhythms in mammals. A: Illustrative traces of neocortical alpha oscillations, sleep spindles, and hippocampal ripples in various species. Note the similarity of frequency, temporal evolution, and waveforms of the respective patterns across species. B: Relationship between brain weight and frequency of the various rhythm classes on a log-log scale. Note the small variation of frequency changes despite increases in brain weight of several orders of magnitude.
Reproduced from Buzsaki et al. (2013).

and cortical localization of sleep spindles are very similar in the mouse and the human brain.

On the one hand, this may not be so surprising. After all, neurotransmitters, their receptors, and the membrane time constants of principal cells and interneurons are also conserved, and these properties underlie various oscillations. Thus, irrespective of brain size, the management of multiple time scales in neuronal networks is supported by the same fundamental mechanisms. On the other hand, the speed of communication between areas varies considerably between small and large brains, making the conservation of rhythms unexpected. For example, for coherent perception of multimodal inputs, the results of local computation in the thalamus and several primary sensory cortical areas should arrive within the integration time window of the target associational cortices. The same applies to the motor side of the brain. As discussed in Chapter 3, the fundamental properties of myosin and actin are largely conserved across mammals. Therefore, the motor command computations in the motor cortex, cerebellum, and basal ganglia should be performed in comparable time windows, and the command signals to the spinal cord should be delivered within the same time range in different species. However, the distances of these structures vary by orders of magnitude across species. Thus, all of the timing constraints required for adequate function have to be reconciled with the complexity imposed by the growing size of the brain. This is not a trivial task, given a 17,000-fold increase of brain volume from the small tree shrew to large-brain cetaceans. The constancy of the many brain oscillations and their cross-frequency coupling effects across species suggest a fundamental role for temporal coordination of neuronal activity.[22]

Conservation of Brain Rhythms Across Species

There appear to be at least two mechanisms that allow scaling of neuronal networks while conserving timing mechanisms. The first mechanism compensates for the increase in neuronal numbers and the enormous numbers of possible connections by shortening the *synaptic path length* between neurons, defined as the average number of monosynaptic connections in the shortest path between two neurons. This problem is akin to connecting cities with highways and airlines to obtain an efficient compromise between the length of the connections and the number of intermediate cities required to get from city A to city B. Directly connecting everything with everything else is not an option in most real-world networks as the number of connections increases much more rapidly than the number of nodes to be connected. An efficient compromise can be achieved by inserting a smallish number of long-range short-cut connections into local connections. The resulting “small-world-like” architectural solution allows for scaling while keeping the average synaptic path length similar as brains increase in size.

While such scaling prevents excessive volume growth, longer axons increase the travel time of action potentials. This would pose a serious problem for neuronal communication. Thus, a second mechanism is needed to compensate

22. Buzsáki et al. (2013). The Supplementary Material section of this review compares hundreds of papers on various rhythms recorded in a multitude of mammalian species.

for these time delays. This requirement is solved by using larger caliber, better insulated, and more rapidly conducting axons in more complex brains. For example, to achieve interhemispheric gamma band synchrony in the mouse brain, a conduction velocity of 5 m/sec is sufficient. On the other hand, maintaining coherent oscillations in the same frequency range in the human brain, with 70–160 mm between hemispheres, requires much more rapidly conducting axons. Thus, the benefits of timing mechanisms in larger brains can be preserved by adding larger-caliber and more strongly myelinated axons that allow signals to travel longer distances within a similar time window. In the human brain, the great majority of interhemispheric (callosal) axons have small diameters (<0.8 μm) but the thickest 0.1% can exceed 10 μm in diameter. These thick fibers scale best with brain size, whereas the fraction of thinner fibers actually decreases. Although adding some large-diameter axons does increase brain volume and metabolic costs, the resulting volume increase is still considerably smaller than would result from a proportional increase in axon caliber across all neurons. This small added fraction of large-diameter axons may be responsible for the minor variability of interhemispheric conduction delays across species.[23] Although the exact neuron types with large-caliber axons are not known, experimental data indicate that at least some them originate from long-range inhibitory neurons. In turn, theoretical considerations and modeling suggest that long-range interneurons are critical for brain-wide synchronization of gamma and potentially other oscillations.[24]

In summary, preservation of brain rhythms in increasingly larger brains can be taken as supporting evidence for the fundamental importance of timing in brain performance. As the various nested rhythms occur in parallel in multiple brain systems, it is clear that oscillations per se do not serve special biological functions. The meaning of gamma oscillations in the sensory olfactory bulb is different from those in prefrontal circuits serving cognitive functions.

23. Aboitiz et al. (2003); Wang et al. (2008). See further discussion on axon diameter in Chapter 12.

24. Densely connected local interneuron networks are supplemented by a small fraction of long-range interneuronal connections, which effectively reduce the synaptic path lengths between distant circuits (Buzsáki et al., 2004) and allow the entrainment of oscillations across distant networks. Long-range interneurons tend to have large-diameter, myelinated axons and can, therefore, provide fast conduits (Jinno et al., 2007). In the brain, most neuronal connections are local but interspersed with long-range shortcuts (Bullmore and Sporns, 2009). This architectural design is reminiscent of mathematically defined "small-world" networks (Watts and Strogatz, 1988). Small-world type networks allow activity to propagate from one neuron (or more likely assemblies of neurons; "node") to distant neurons. Steve Strogatz's book (*Sync*, 2003) is an excellent reference for small-world networks, a term he coined. For connection rules of the brain, see Chapter 12.

Instead, the benefits of a particular oscillation depend on the function of the brain system that supports it.

ENTRAINMENT OF OSCILLATORS: TRACKING TEMPORAL DYNAMICS OF SPEECH

When two oscillators with identical frequency engage each other, the outcome depends on the phase of the two rhythms. In-phase interactions induce resonance and, as a result, amplification.[25] In contrast, opposing phase interactions may annihilate or dampen the rhythm. Oscillators with noninteger relationships induce perpetual interference. This is typical of brain rhythms, and the interference mechanism explains why brain dynamics is constantly changing, similar to the interference of ocean waves. Occasionally, the oscillatory reader mechanism may transiently adjust its phase to the incoming inputs. Such phase adjustment is among the most important flexible features of brain oscillators. This is similar to how musicians in an orchestra keep the beat. If the first violinist is a bit faster, the rest of the musicians adjust the timing of their movements. Furthermore, the phase separation of the maximum and minimum spiking activity of the information-transmitting principal cells make oscillators a natural parsing mechanism. As mentioned earlier, this is an effective separating mechanism of neuronal messages, a fundamental requirement for syntactical operations.[26]

Speech Rhythms Are Brain Rhythms

A striking example of the selection and amplifying properties of brain oscillators is their responses to speech. Rhythms of human speech are remarkably similar

25. Single neurons can also exploit resonance and filtering. The leak conductance and capacitance of the neuronal membrane are mainly responsible for the low-pass filtering of neurons (essentially an RC filter). In contrast, several voltage-gated currents, whose activation range is close to the resting membrane potential, act as high-pass filters, making the neuron more sensitive to fast trains of spikes. These resonant-oscillatory features allow neurons to select inputs based on their frequency. Neurons with high-pass and low-pass filtering can be combined to build neuronal networks that can function as band-pass resonators (Llinás et al., 1988; Alonso and Llinás, 1989; Hutcheon and Yarom, 2000). Cortical interneuron classes have a wide range of preferred frequencies (Thomson and West, 2003), and their diverse frequency-tuning properties are important for setting network dynamics.

26. The volume edited by Bickerton and Szathmáry (2009) contains many excellent chapters on syntactical rules related to both language and neuronal activity.

across all spoken languages, and our brains are tuned to efficiently track and extract such information. We can easily recognize a stutterer or a person with a speech impairment in any language because of the altered speech rhythm. The so-called *prosodic features* of speech, such as intonation, stress, and pause, are characteristic of individuals, yet they also share common features among all people, varying between 0.3/s and 2/s (delta band).[27] Syllables also repeat somewhat rhythmically between 4 and 8 times per second (theta band), while phonemes and fast transitions are characterized by a frequency band between 30/s and 80/s (gamma band). Two features of brain oscillations can facilitate the extraction of hierarchically organized continuous speech. First is the correspondence between speech and native brain rhythm frequencies. Their cross-frequency phase coupling can amplify sound features and assist with the segmentation of speech components. Second, the ability of neuronal oscillators to reset their phase can effectively track the temporal features of quasi-rhythmic spoken language.

It is tempting to draw a parallel between neuronal syntax, supported by brain rhythms, and language syntax. Brain rhythms may constrain the way our motor system controls utterance of sound. In return, the "phonetical syntax" of speech, referring to the sequences of sounds, can influence brain oscillations. I make this comparison later with the explicit acknowledgment that there is a difference between speech syntax and syntactic coding of speech. Yet the brain-constructed and human community-constructed syntactic rules often seem correlated.

Experiments using EEG, ECoG, or MEG methods have demonstrated that the edges of the sound envelope of speech can reset the phase of slow brain oscillations in the right temporal and frontal areas and that speech sound envelope fluctuations are correlated with amplitude variations in the delta band. The time-varying dynamics of syllable sequences can be faithfully tracked by the amplitude of theta oscillations and the integrated amplitude of gamma activity. In high-density ECoG recordings from the surface of the superior temporal gyrus (nonprimary auditory cortex) of patients, the spectrotemporal auditory features of continuous speech could be reliably reconstructed from neuronal signals, and many words and pseudowords could be statistically

27. Prosodic features describe the auditory qualities of speech beyond individual phonemes. Prosodic phrases are concatenated phonemes, syllables, and morphemes, reflecting suprasegmental features of speech. They are characterized by the pitch and intonation of voice, rhythm, stress, and loudness. A *phoneme* is an abstract sound feature that differentiates between two words (e.g., "house" vs. "mouse"). A *syllable* is a unit of pronunciation with one vowel. A *morpheme* is a word or part of a word and is the smallest grammatical unit that has meaning. For example, information [in-for-ma-tion] has four syllables and one morpheme.

separated. Similarly, when song segments were played to the research participant, the characteristic time-varying sound patterns could be recognized from the smoothed power changes of high gamma oscillations. The requirement for tracking the temporal dynamic of speech by brain rhythms is illustrated by experiments in which test sentences are time-warped. When speech is compressed, but comprehension is maintained, both phase-locking and amplitude-matching between the temporal envelopes of speech and the recorded brain signal are maintained. However, when speech compression is large, brain oscillations no longer effectively track speech, and comprehension is degraded.[28]

An outside-in argument could be that the dynamic relationship between speech and brain oscillations is due to the imposition of the speech patterns on brain patterns. In other words, speech stimuli "train" the brain to effectively segment and parse speech content.[29] This argument has little validity because neuronal oscillations are the same in all mammals, and speech rhythms are the same in every human culture. Instead, I suggest that the coding strategy of the auditory system is matched to internally organized brain rhythms. In support of this view, when rats hear complex sounds, such as music or broadband pink noise, thalamocortical patterns in the superficial layers of the auditory cortex are also segmented into 2–4 frames per second. As in humans, these events reliably reset the phases of the local field potentials at specific times of the acoustic stimulus. Overall, the preservation of brain rhythms across species suggests that human speech was built on preexisting brain dynamics.

28. An important background paper is Shannon et al. (1995), which demonstrated that speech pattern recognition, in general, can use both spectral and temporal cues. Several works have suggested, implicitly or explicitly, that syntactical rules of language are related to neural syntax (Buzsáki, 2010; chapters in the edited volume by Bickerton and Szathmáry, 2009, also discuss such suspected links). For physiological support for the role of brain oscillations in segmentation of speech, see Ahissar et al. (2001); Lakatos et al. (2005, 2008); Howard and Poeppel (2010); Ding and Simon (2012). Valuable ECoG data are typically obtained from patients with brain tumors or epilepsy using subdural electrode arrays, when patients give their informed consent for such tests (Pasley et al., 2012; Sturm et al., 2014).

29. The probability distributions of the amplitude envelope and the time-frequency correlations of all natural sounds are quite similar. Animal vocalizations and human speech are further characterized by the low temporal modulation of most spectral power. The distribution of the amplitude envelopes exhibits characteristic shapes for natural sounds and a relatively uniform distribution for the log of the amplitudes for vocalizations (characterized by the 1/f form, as is the case for the EEG signal). Largely for these reasons, Singh and Theunissen (2003) suggested that the auditory system has evolved to process behaviorally relevant sounds. Under this hypothesis, the statistics of the spectrotemporal amplitude envelopes of the sound should be critical for auditory brain areas to extract sound identity.

Syntactic Segmentation, Grouping, and Parsing by the Brain's Oscillatory System

So far, we have discussed that brain rhythms are efficient for segmenting and parsing continuous natural sounds and that matching between the modulation capacities of the auditory cortex and speech dynamics is a prerequisite for comprehension (although certainly not sufficient). However, the brain's response to the sound envelope is not simply a matter of oscillatory coupling but also involves feature extraction as well. In support of this idea, the magnitude of theta-gamma phase coupling is stronger when the listener understands the speech compared to when the same speech segment is played backward or when the speech components are randomly shuffled.[30] In other words, the variation of cross-frequency coupling magnitude may carry the meaning.

Brain Rhythms at a Cocktail Party

At a cocktail party, where several people may speak at the same time, the speech system helps us by grouping the complex auditory scene into separate "objects." Intelligible speech is extracted by phase-locking the brain's rhythms to the speech of a chosen person. This is a pertinent example of selective gain control (Chapter 11) because the sound emitted by one individual (i.e., the target auditory object) is amplified by phase-resetting and resonance. At the same time, the speech streams from other speakers are filtered and suppressed

30. Functional magnetic resonance imaging (fMRI) studies indirectly support the EEG/MEG data. The blood oxygen level-dependent (BOLD) signal in the auditory cortex increases irrespective of whether intelligent or unintelligent (scrambled) speech is presented. In temporal speech areas (such as Broca's area; Brodmann areas 45, 47), coherent information at the sentence level is needed to induced blood flow changes, whereas parietal and frontal areas (Brodmann areas 39, 40, 7, and 22) respond only when intact paragraphs of intelligent speech are presented (Lerner et al., 2011). Two major language-relevant systems can be distinguished in the brain: a dorsal and a ventral system. The dorsal system involves Broca's area (in particular Brodmann area 44), which, together with the posterior temporal cortex, supports hierarchical syntactic computation and comprehension of complex sentences. The ventral system (area 45/47) and the temporal cortex support the processing of lexical-semantic and conceptual information (see e.g., Hagoort, 2005; Berwick et al., 2013). Neurons in the temporal lobe can respond to semantic categories or can be activated by the photograph, sketch, or even name of a familiar person, implying a high degree of semantic abstraction (Quian Quiroga et al., 2005). The speed of word naming depends on "semantic richness." Neuronal assemblies can be combined for many word representations, with semantically related words being encoded by hypothetically overlapping neural ensembles (Li et al., 2006; Sajin and Connine, 2014; Friederici and Singer, 2015).

because their speech arrived during the "ignoring" phase of the rhythms of the listener's brain.

In a real cocktail party, speakers are at physically distinct locations, so you might argue that their spatial position can be triangulated by binaural listening.[31] However, the contribution of spatial localization mechanisms can be excluded in laboratory settings when listeners are asked to selectively attend to one of two competing speakers (typically male and female) heard through a single earphone. Such experiments show that the selection of a speaker relies on the match between the shape of the frequency and the temporal modulation of the chosen speaker's sound and the listener's low-frequency (delta, theta band) MEG/EEG activity. This selective extraction mechanism is achieved primarily by phase-tuning the neuronal oscillators to the prosodic and syllabic tempo of the preferred speaker because varying the sound intensity of the competing speaker, within a reasonable range, does not affect comprehension.

These observations, based on healthy human research participants using noninvasive recording methods (MEG and EEG), are supplemented by high-resolution multielectrode surface recordings (ECoG) from the posterior superior temporal gyrus (a higher order auditory area) in patients with epilepsy. Analysis of the speech spectrograms of a mixture of speakers reconstructed from the envelope of high gamma rhythm (75–150/s) shows that the spectral and temporal features of the ECoG signal triggered by the sound of the attended speaker are more salient than those of induced by competing speakers. These findings demonstrate that not only acoustic features but also intentional speaker selection can be improved by phase-locked oscillations.[32]

31. Localization of the sound source also uses timing information. Sounds from a speaker on your right arrive at the right ear first and, after a slight delay, at the left ear. This time difference may be shorter than the duration of a single action potential in the auditory nerve (less than 1 ms). Of course, slight differences in sound intensity are also present between the two ears. Mark Konishi and colleagues at the California Institute of Technology showed that *medial superior olive* neurons in barn owls are selective for timing differences. These neurons are "coincidence detectors": some neurons respond selectively at a 20 μsec time lag, others at 50, 75, 100 μsec, and so on. Their spike output therefore signals the direction of the sound source. Sound intensity differences between the two ears may also assist in computing the distance to the source (Knudsen and Konishi, 1978).

32. Acoustic and semantic features of words are most often assumed to be independent, even though many linguists, starting perhaps with Socrates, have speculated that "good" words have sounds that suit their meaning. Recent large-scale studies rekindled this old debate by demonstrating such a link in several languages. Not only sounds imitating those made by animals (such as "moo") are similar across languages. When people are asked whether "bouba" or "kiki" is the right word for an amoeboid or star-shaped (spiky) figure, speakers of different languages consistently make the right guess (they choose kiki). The word used to denote "sand" often contains the sound /s/ examined across 4,000-plus languages. The vowel /i/ often refers

A computational model (an artificial classifier) trained solely on examples of one of the speakers could recognize both attended words and speaker identity despite the interfering effects of the competing speaker. When behavioral errors occurred, they were reflected in the degradation of oscillatory tuning. Overall, cortical activity does not merely reflect responses to acoustic stimuli but, in addition, it also can identify more complex aspects of speech, including the listener's goal.[33]

In this chapter, we examined how neurons and cell assemblies are governed by their collective behavior, as expressed by brain oscillations. Now it is time to learn how brain oscillations assist with organizing cell assemblies into longer sequences and the functions they support. Curious? Turn the page to Chapter 7.

SUMMARY

From the analogy between speech syntax and brain rhythms, I made the assumption that the hierarchy of brain oscillations may parse and group neuronal activity to decompose and package neuronal information in communication between brain areas. Because all neuronal oscillations are based on inhibition, they can parse and concatenate neuronal messages, a prerequisite for any coding mechanism. The hierarchical nature of cross-frequency-coupled rhythms can serve as a scaffold for combining neuronal letters into words and words into sentences. Brain oscillations, present in the same form in all mammals, represent a fundamental aspect of neuronal computation, including the generation of movement patterns, speech, and likely, music production. Neuronal oscillators readily entrain each other, making the exchange of messages between brain areas effective. I speculate that the roots of language and musical syntax emanate from this native neural syntax since the same brain rhythms that assist

to small size, /r/ to roundness, and /m/ to mother or breast even in nonrelated languages (Blasi et al., 2016; Fitch, 2016).

33. Schroeder and Lakatos (2009); Ding and Simon (2012); Mesgarani and Chang (2012); Zion Golumbic et al. (2013). Other investigators emphasize the role of low-frequency oscillators in speech segmentation. Scalp EEG recordings suggest that 4–8 Hz activity in auditory cortex is important for tuning in to the continuous speech content of an attended talker. Interhemispheric asymmetry of alpha power (8–12 Hz) at parietal sites can indicate the direction of auditory attention to a speaker (Kerlin et al., 2010). While the brain efficiently solves the auditory object segregation problem, it still remains a major challenge for automatic speech recognition algorithms (Cooke et al., 2010).

in the generation of speech and music are also responsible for their syntactical segmentation and integration.[34]

Brain waves detected in either LFPs or EEG signals can reveal certain aspects of signal transformation because the experimenter has access to both an external event (speech) and an internal event (brain oscillations). However, because the brain does not use LFPs or EEG for communication,[35] it remains to be demonstrated whether and how brain networks use such packaging mechanisms for their own purposes. The semantic content of information packages can be read out from neuronal spikes alone, but to become information, spike patterns need syntactical rules known to both sender and receiver mechanisms. Brain oscillations represent a candidate cipher because they are present in all the brain regions of all communicating parties.

34. Singh and Theunissen (2003); Buzsáki (2010); Pulvermüller (2010); Giraud and Poeppel (2012).

35. There are exceptions. Electric fields generated by synchronous activity of nearby neurons can affect the membrane potential of the same neurons and increase synchrony further. Such local "ephaptic" effects are especially effective in structures with high neuronal density and regular architectures, such as the hippocampus (Anastassiou et al., 2010).